

Simulation report

X-ray imaging of pigments on wood

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Purposes

The main purpose of this work is to simulate an approximate X-ray imaging output when applying such technique to paintings containing metal-based pigments of cultural interest. A wood canvas and inorganic matrices mixed up with pure pigments are taken into account, and source and detector characteristics are modeled on the actual source and detector that are supposed to be used for measurements further on.

Simulations and analyses are carried out using a Graphical User Interface developed from the SpekPy Python library and employing the ImageJ software.

Configuration

Source

Source configuration:

- kV peak = 100 *kV*
- current = 0.2 *mA*
- exposure time = 120 *s*
- filtration = *Al*, 0.2 *mm* width
- distance between source and sample = 100 *cm*
- including air attenuation
- anode material and geometry: *W*, transmission
- anodic angle = -90.0°
- target thickness = 0.5 μm

Source statistics:

- Mean energy = 35.902 *keV*
- Incident fluence = 30257429.609 photons/ mm^2 (total number of incident photons on a surface of 1 mm^2 , during the whole exposure time)
- Incident photon flux = 252145.247 photons/ $\text{s} \cdot \text{mm}^2$ (incident fluence per unit time)

Sample

Background:

Air, predefined: density = 0.001205 g/cm^3 , composition = N_2 , thickness = 5.0 *mm*

Simulation of a real painting:

- woody support, expected heterogeneous (i.e. an average density over many possible wood types is considered)
- white lead layer deposited on wood to make its surface homogeneous, ready to be painted on
- different pigments with metallic components: copper-based (Egyptian Blue, Malachite, Azurite), lead-based (Minium, Massicot), iron-based (Iron Oxide Yellow, Dark Ochre, Hematite), mercury-based (Cinnabar)

Hence the details (custom materials) in two different simulations:

Detail	Sample name	Density [g/cm^3]	Composition	Shape	Center (x_c, y_c) [mm]	Size (x, y) [mm]	Thickness [mm]
1	Wood	0.75	60% $C_6H_{10}O_5$ 14% $C_9H_8O_2$ 13% $C_{10}H_{12}O_3$ 13% $C_{11}H_{14}O_4$	box	(0, 0)	(23, 23)	50.0
2	“Biacca”	6.94	$2PbCO_3 \cdot Pb(OH)_2$	box	(0, 0)	(20, 20)	0.4
3	Egyptian Blue	3.10	$CaCuSi_4O_{10}$	box	(6, -7)	(6, 6)	0.2
4	Malachite	3.80	$CuCO_3 \cdot Cu(OH)_2$	box	(8, -4)	(4, 4)	0.2
5	Azurite	3.90	$2CuCO_3 \cdot Cu(OH)_2$	box	(5, -4)	(6, 4)	0.2
6	Minium	8.70	Pb_3O_4	box	(-6, -6)	(7, 7)	0.2
7	Massicot	9.50	PbO	box	(-2, -4)	(6, 7)	0.2
8	Iron Oxide Yellow	3.40	40% $FeO(OH)$ 20% Al_2O_3 20% SiO_4 20% $CaCO_3$	box	(-7, 7)	(5, 5)	0.2
9	Dark Ochre	2.90	22% $Fe_2O_3H_2O$ 48% $CaCO_3$ 30% $CaSO_4$	box	(-4, 4)	(6, 3)	0.2
10	Hematite	4.60	80% Fe_2O_3 13% SiO_2 7% Al_2O_3	box	(7, 7)	(5, 6)	0.2
11	Cinnabar	8.10	HgS	box	(6, 3)	(5, 7)	0.2

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1	Wood	0.75	60% $C_6H_{10}O_5$ 14% $C_9H_8O_2$ 13% $C_{10}H_{12}O_3$ 13% $C_{11}H_{14}O_4$	box	(0, 0)	(23, 23)	50.0
2	“Biacca”	6.94	$2PbCO_3 \cdot Pb(OH)_2$	box	(0, 0)	(20, 20)	0.1
3	Lead Yellow	8.80	$PbSnO_4$	box	(-2.5, -3.0)	(7, 5)	0.2
4	Naples Yellow	6.50	$Pb_2Sb_2O_7$	box	(-5.0, -4.5)	(6, 8)	0.2
5	Orpiment	3.48	As_2S_3	box	(2.5, -3.0)	(6, 5)	0.2
6	Realgar	3.56	As_4S_4	box	(5, -3)	(4, 7)	0.2
7	Cadmium Yellow	4.82	CdS	box	(5.0, 3.5)	(4, 8)	0.2
8	Cadmium Red	5.25	$CdS \cdot CdSe$	box	(2.5, 4.0)	(5, 5)	0.2
9	“Smaltino”	4.10	$CoO \cdot SiO_2$	box	(-5, 5)	(5, 5)	0.2
10	Cobalt Blue	4.30	$CoO \cdot Al_2O_3$	box	(-6.5, 2.0)	(5, 7)	0.2
11	Cobalt Violet	3.70	$Co_3(PO_4)_2$	box	(-2.5, 5.0)	(6, 7)	0.2

Detector

- sensor material = $CdTe$
- sensor thickness = 1.0 mm
- pixel size = 0.055 mm
- image width = 448 pixels
- image height = 512 pixels

To first order, the detector efficiency is assumed to be constant (independent of energy) and it is set to 0.75, and the detector resolution is optimistically set to $FWHM = 4.0 \text{ keV}$.

Consequently, the energy bin width is set to $FWHM/2 = 2 \text{ keV}$ (i.e. 50 energy bins, from stack 1 corresponding to $E = 1 \text{ keV}$ to stack 50 corresponding to $E = 99 \text{ keV}$).

Source statistics:

- Expected background fluence = 22142981.471 photons/ mm^2 (total number of transmitted photons by a background surface of 1 mm^2 , during the whole exposure time)
- Total background counts = 16607236.063 photons/ mm^2 (total number of detected photons on a background surface of 1 mm^2 , during the whole exposure time)

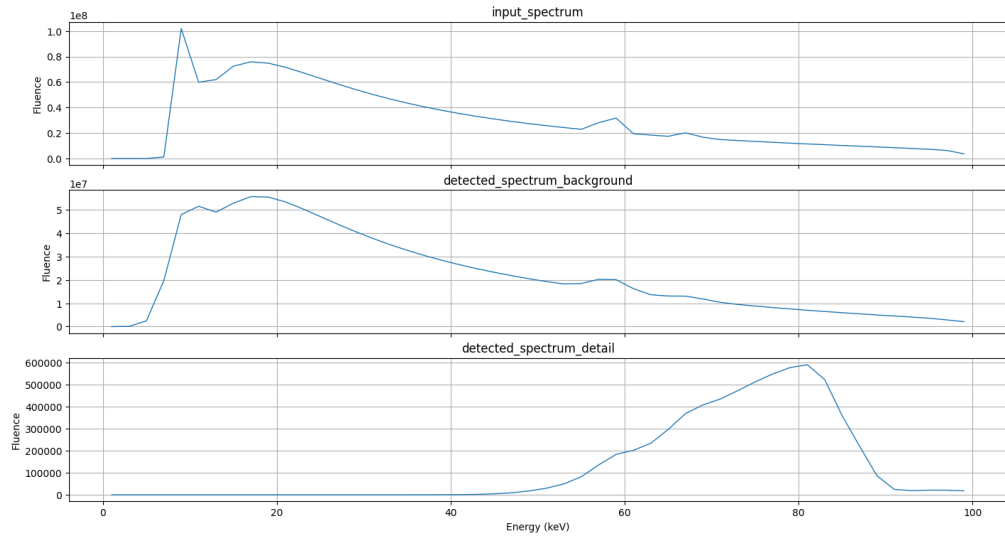
- Expected detail fluence = $129220.903 \text{ photons/mm}^2$ (total number of transmitted photons by an average sample surface of 1 mm^2 , during the whole exposure time)
- Total detail counts = $96915.677 \text{ photons/mm}^2$ (total number of detected photons on an average sample surface of 1 mm^2 , during the whole exposure time)

The expected background and detail fluence (i.e. the ideal total background and detail counts) are computed starting from the Lambert-Beer law, the incident fluence, the linear attenuation coefficient of considered materials (which depends on the incident photon energy) and their predefined thicknesses.

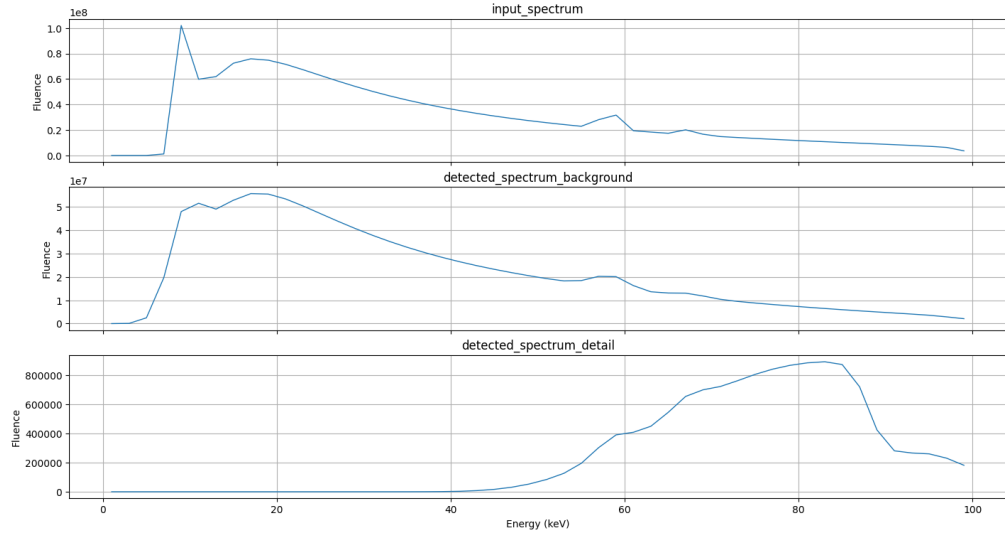
The expected background and detail counts are obtained multiplying the respective ideal counts by the constant efficiency (actual counts would result from the convolution of the energy-dependent detector efficiency and the expected energy-dependent fluences, stack by stack over the whole energy spectrum).

Spectra

Spectra are plotted starting from the obtained `.csv` files of first and second simulation:



(a) First simulation



(b) Second simulation

Figure 1: Global spectra: the input spectrum is generated from the source configuration, the background and detail spectra display the fluence, i.e. the total ideal number of detected photons per cm^2 and keV , as a function of energy

The detected background spectrum resembles the input spectrum, as X-rays are negligibly absorbed by air. The detail-transmitted spectrum shows a strong absorption of low-energy X-rays (because of 5 cm-wide wood and most of all because of white lead almost covering the whole area), and increasing X-ray energies are gradually less absorbed. A transmittance peak occurs at an energy value $\in [80 \text{ keV}, 90 \text{ keV}]$, then transmittance decreases rapidly: this effect can be attributed to the K-edges of heavy elements, in particular lead and (in first simulation) mercury present in pigments (respectively, $K_{Pb} \simeq 88 \text{ keV}$ and $K_{Hg} \simeq 83 \text{ keV}$, which seem to be smoothed in a single continuous fluence decrease because of limited resolution).

Definition of SNR and CNR

To implement an X-ray imaging with sufficient quality and detail visibility, it is crucial to determine quantitative parameters such as Signal to Noise Ratio and Contrast to Noise Ratio.

SNR and CNR on counts

Since counts follow the poissonian statistics, it is reasonable to assume, in a transmittance region (i.e. at energies that lead to a mean number of counts sensitively different from zero), that the photons signal to noise ratio is:

$$SNR = \frac{\langle N \rangle}{\sigma_N} \simeq \frac{N}{\sqrt{N}} = \sqrt{N}$$

A value $SNR > 5$ (Rose threshold) should allow the human eye to distinguish some foreground structure from the background and a value $SNR > 10$ should be sufficient to identify edges and macro-structures, but a $SNR \sim 100$ should be reached in order to achieve a high-resolution radiographic imaging.

An important related parameter is the contrast to noise ratio, which can be computed starting from counts on a Region Of Interest and on a background (in this case, wood or white lead) as:

$$CNR = \frac{|\langle N \rangle_{ROI} - \langle N \rangle_{backg}|}{\sqrt{\sigma_{ROI}^2 + \sigma_{backg}^2}} \simeq \frac{|\langle N \rangle_{ROI} - \langle N \rangle_{backg}|}{\sigma_{backg}} \simeq \frac{|N_{ROI} - N_{backg}|}{\sqrt{N_{backg}}}$$

A value $CNR > 5$ should be sufficient to distinguish between details and background, but a $CNR > 10$ is desirable for a high-quality imaging.

SNR on total attenuation

It might be useful to compute the Signal to Noise Ratio on total attenuation data $A = -\ln(I/I_0)$: noise values can be obtained via error propagation on function A , assuming I and I_0 to be independent variables:

$$\sigma_A = \sqrt{\left(\frac{\partial A}{\partial I} \cdot \sigma_I\right)^2 + \left(\frac{\partial A}{\partial I_0} \cdot \sigma_{I_0}\right)^2} = \sqrt{\left(\frac{\sigma_I}{I}\right)^2 + \left(\frac{\sigma_{I_0}}{I_0}\right)^2} \simeq \sqrt{\frac{1}{I} + \frac{1}{I_0}} \Rightarrow SNR = \frac{\langle A \rangle}{\sigma_A}$$

Output image analysis: counts

Each stack (i.e. Z slice) is an energy bin; when Z-axis profile is plotted, the vertical axis displays mean photon counts per pixel and energy bin (corresponding to the photon fluence multiplied by the pixel area), sampled from a Poissonian, the horizontal axis displays the slice.

Wood

Wood background appears to be almost transparent to X-rays, because of its organic composition.

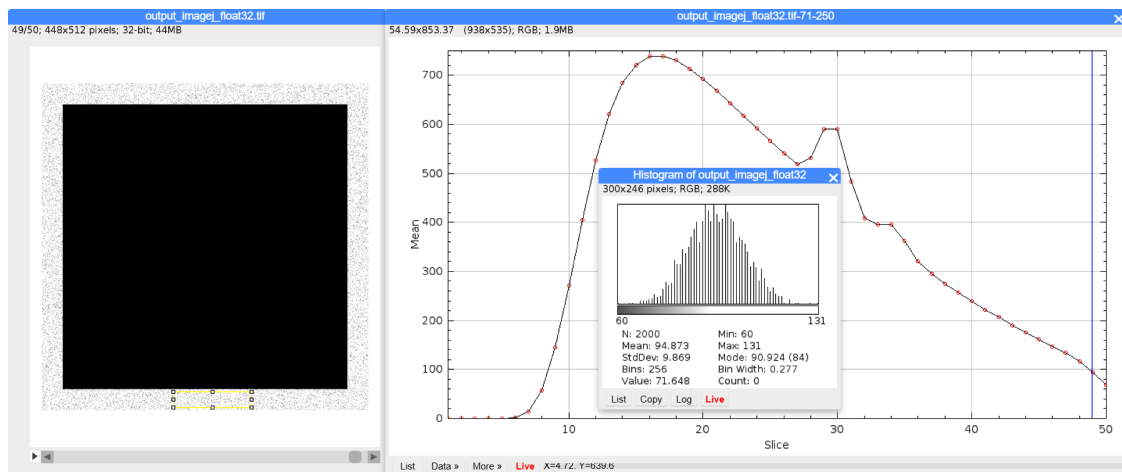


Figure 2: Counts on a wood ROI and respective histogram, which shows a $SNR = 9.613$

White lead

White lead might influence the imaging quality because of the lead's K-edge.

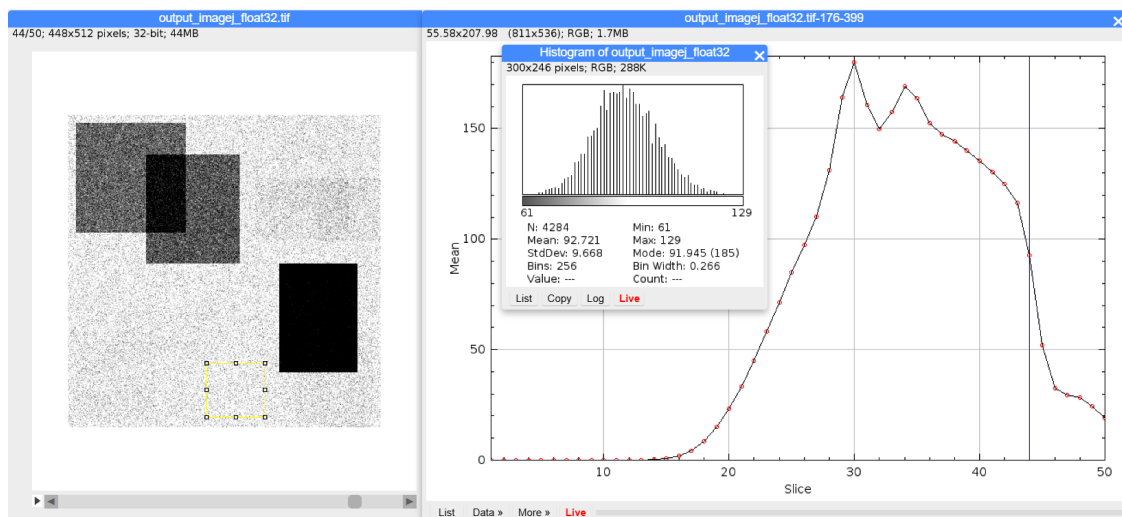
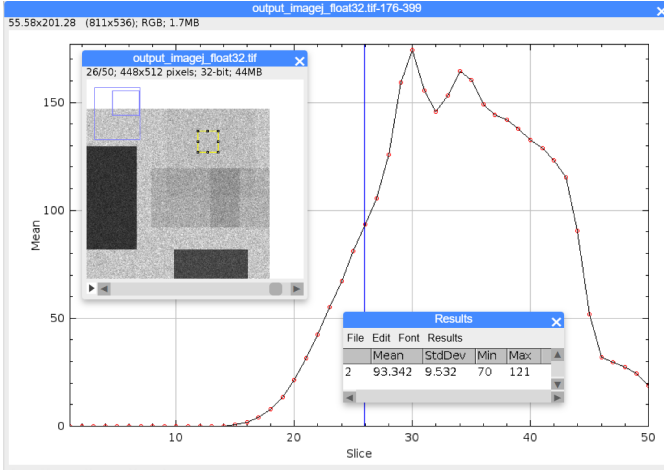


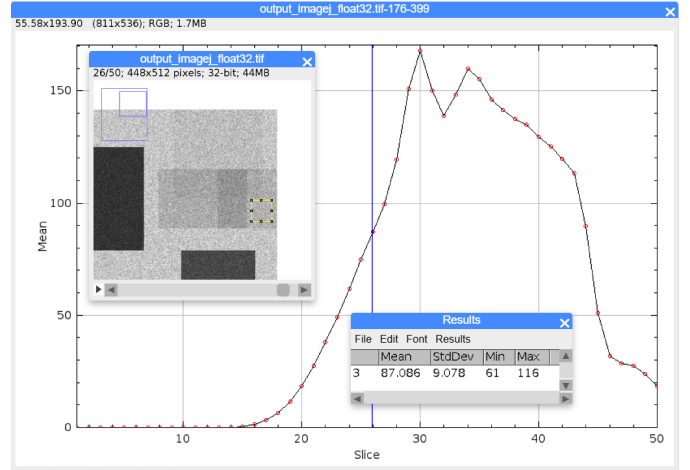
Figure 3: Counts on a “Biacca” ROI and respective histogram, which shows a $SNR = 9.591$

Copper-based pigments

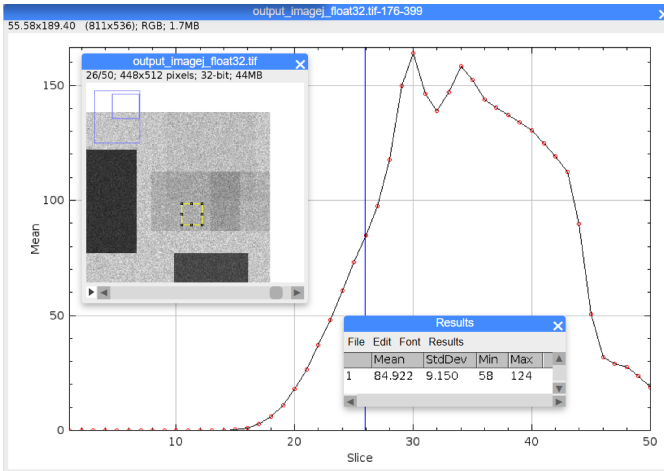
Egyptian blue, Malachite and Azurite all display the same behaviour, hence they cannot be identified univocally. They can be distinguished from wood but they are hardly distinguishable from a possible white lead background.



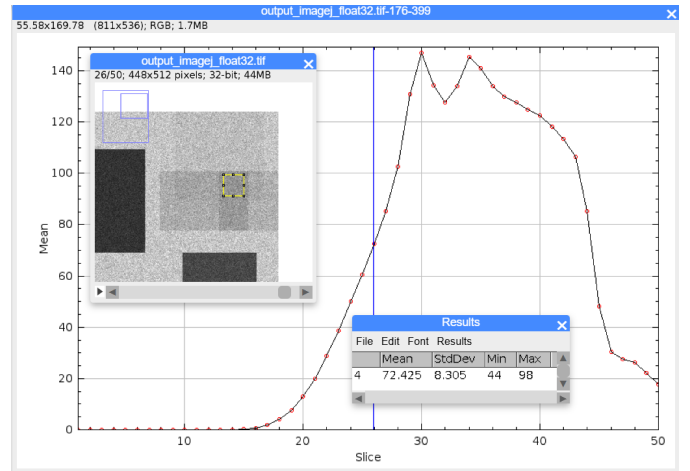
(a) Counts on an Egyptian Blue ROI, $SNR = 9.782$



(b) Counts on a Malachite ROI, $SNR = 9.593$



(c) Counts on an Azurite ROI, $SNR = 9.281$

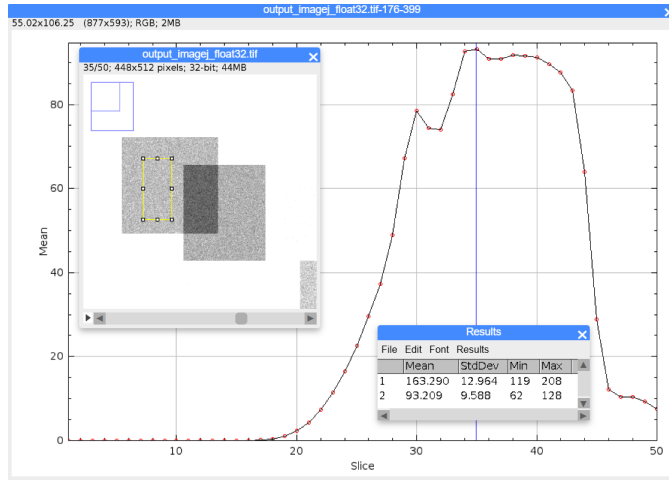


(d) Counts on a ROI of three pigments superposition

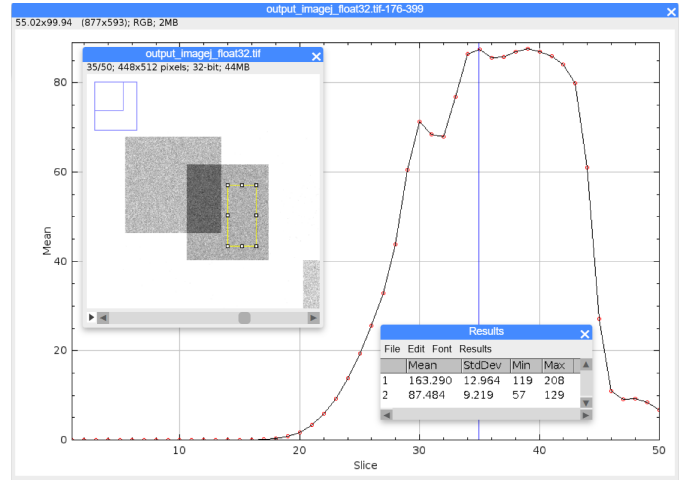
Figure 4: Copper-based pigments: the superposition of the three samples shows a $CNR = 20.947$ with respect to wood and a $CNR = 2.603$ with respect to white lead

Lead-based pigments and Cinnabar

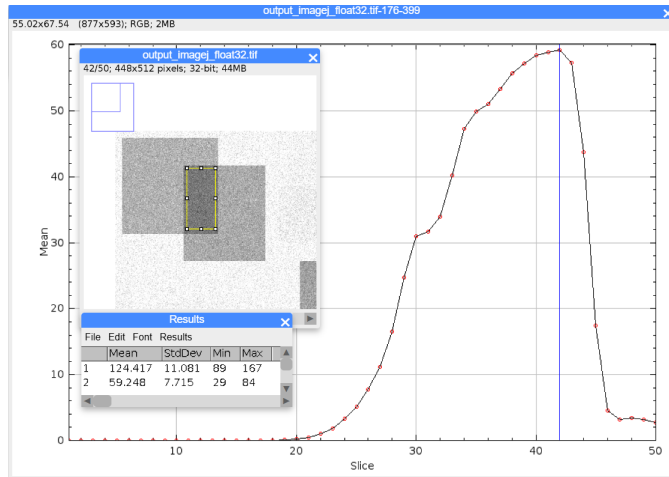
It seems not possible to tell Minium from Massicot but they are easily distinguishable from the background and from other pigments. Lead-based pigments and Cinnabar behave similarly, but they seem to be distinguishable with the set instrumental resolution.



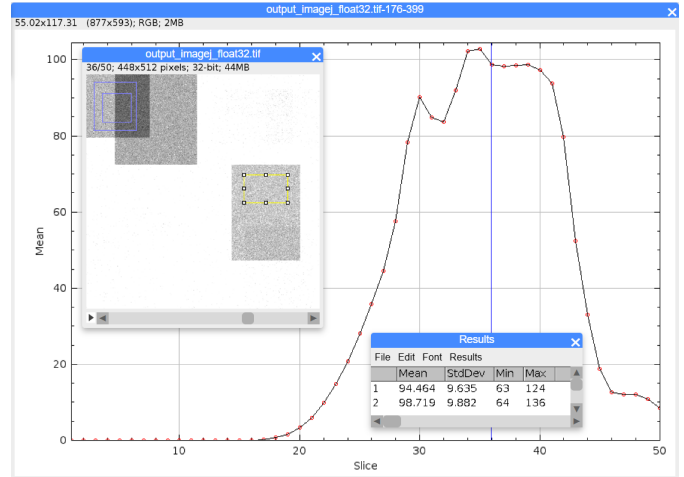
(a) Counts on a Minium ROI, $SNR = 9.721$ and $CNR = 5,406$ with respect to the white lead background



(b) Counts on a Massicot ROI, $SNR = 9.490$ and $CNR = 5,847$ with respect to the white lead background



(c) Counts on a ROI of Minium and Massicot superposition, $CNR = 5.881$ with respect to white lead



(d) Counts on a Cinnabar ROI, $SNR = 9.804$, predominating over a superimposed iron-based red pigment ($CNR = 0.431$)

Figure 5: Lead-based pigments and mercury-based pigment

Similarly, Lead Yellow and Naples Yellow cannot be separated only using counts, although they evidently stand out from the white lead background. Moreover, the presence of tin in Lead Yellow and the presence of antimony in Naples Yellow cannot be spot in counts images.

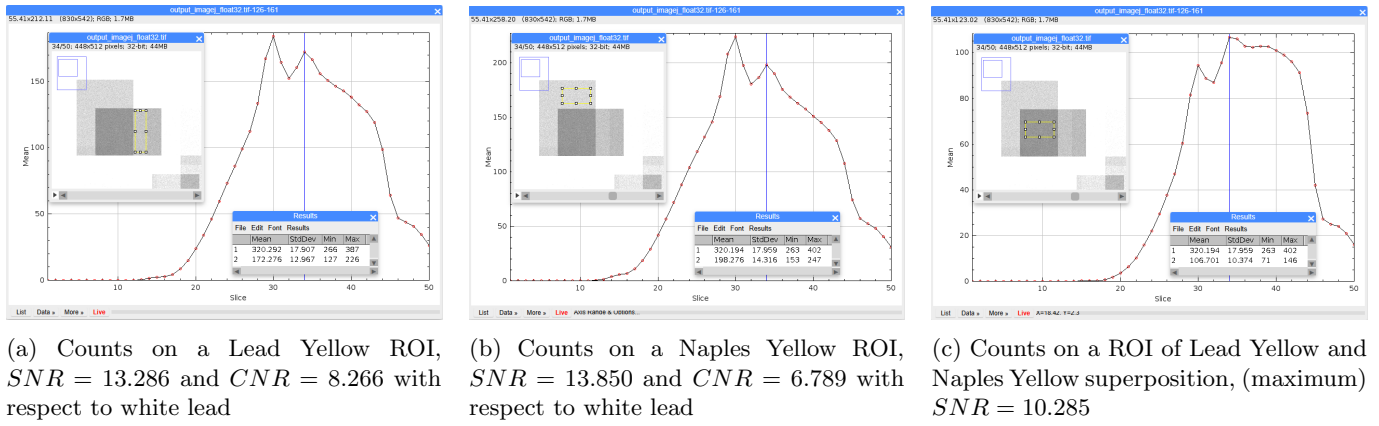


Figure 6: Lead-based yellow pigments, also containing tin and antimony

Iron-based pigments

Iron-based pigments behave similarly to copper-based pigments: they are hardly identifiable and hardly distinguishable from the white lead background (unlike wood background).

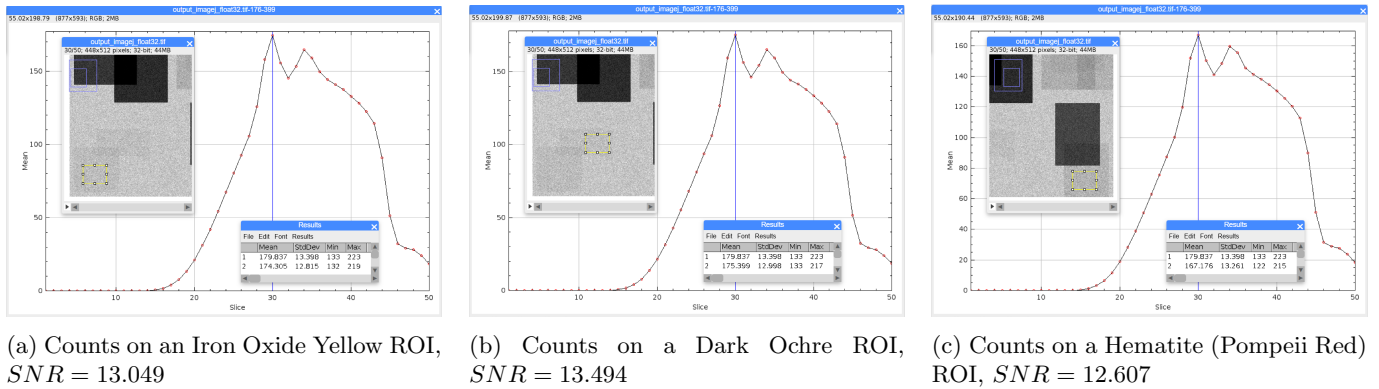
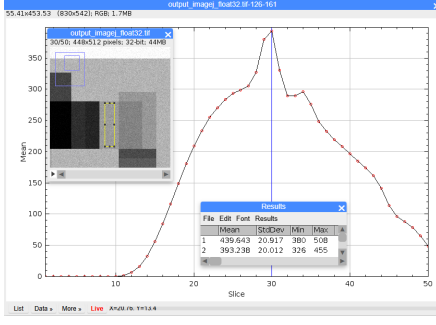


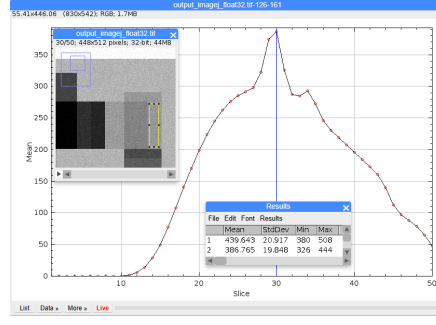
Figure 7: Iron-based yellow and red pigments; visibility is determined by the lead carbonate matrix: the three pigments show a $CNR < 1$ with respect to the white lead background

Arsenic-based pigments

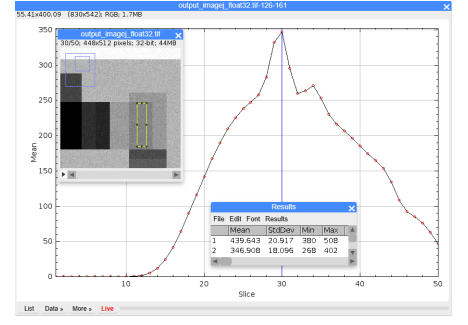
Orpiment and Realgar display an analogous behaviour, as they are characterised by a similar composition and density, and they can be distinguished from white lead as long as it has a small thickness (in this simulation, even smaller than the pigments' thickness).



(a) Counts on an Orpiment ROI, $SNR = 19.650$ but $CNR = 2.219$ with respect to white lead



(b) Counts on a Realgar ROI, $SNR = 19.486$ but $CNR = 2.528$ with respect to white lead

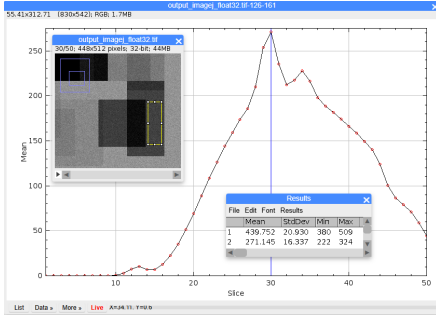


(c) Counts on a ROI of Orpiment and Realgar superposition, (maximum) $SNR = 19.170$

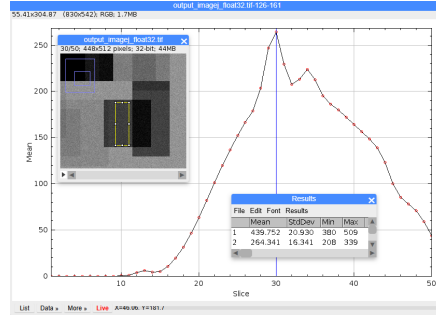
Figure 8: Arsenic-based orange pigments

Cadmium-based pigments

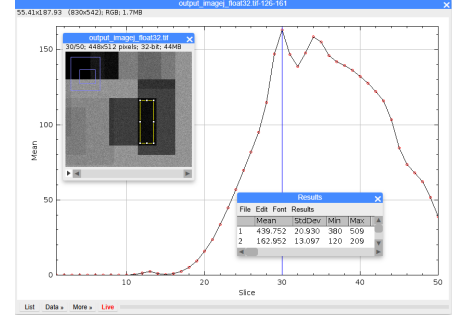
These yellow and red pigments seem to be displaying a weak counts peak around stack 13 (i.e. counts decreasing at $E \approx 27 \text{ keV}$), which should be characteristic of cadmium, but it is expected to be undetectable in actual measurements as $SNR < 5$ in both cases.



(a) Counts on a Cadmium Yellow ROI, $SNR = 16.597$ and $CNR = 8.056$ with respect to white lead



(b) Counts on a Cadmium Red ROI, $SNR = 16.177$ and $CNR = 8.381$ with respect to white lead

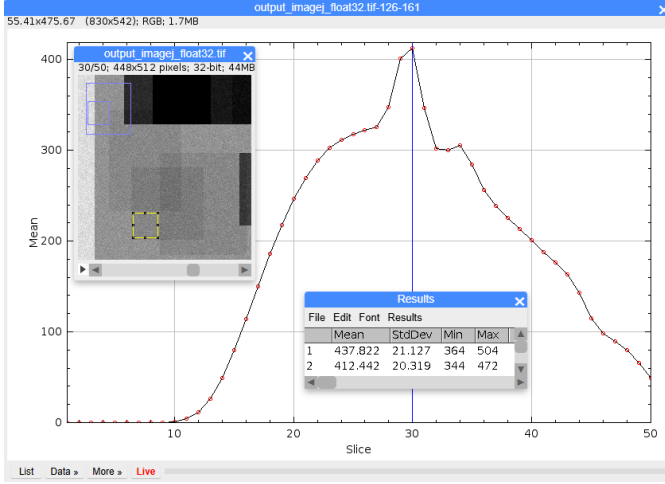


(c) Counts on a ROI of Cadmium Yellow and Red superposition, (maximum) $SNR = 12.442$

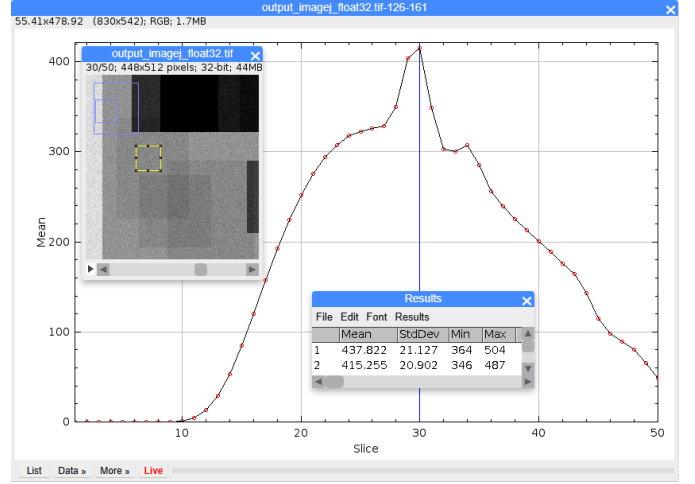
Figure 9: Cadmium-based pigments; counts on a ROI of cadmium-based pigments superimposed to arsenic-based or cobalt-based pigments do not show evidently different results (not shown in the figure)

Cobalt-based pigments

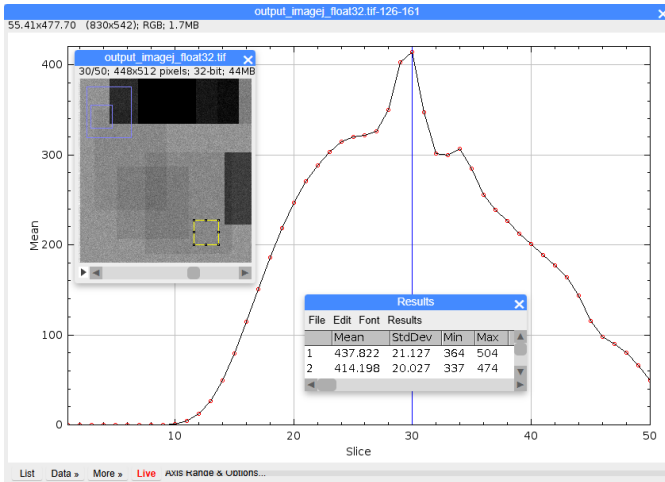
Cobalt-based pigments behave similarly to iron and copper-based pigments, as they are not uniquely identifiable.



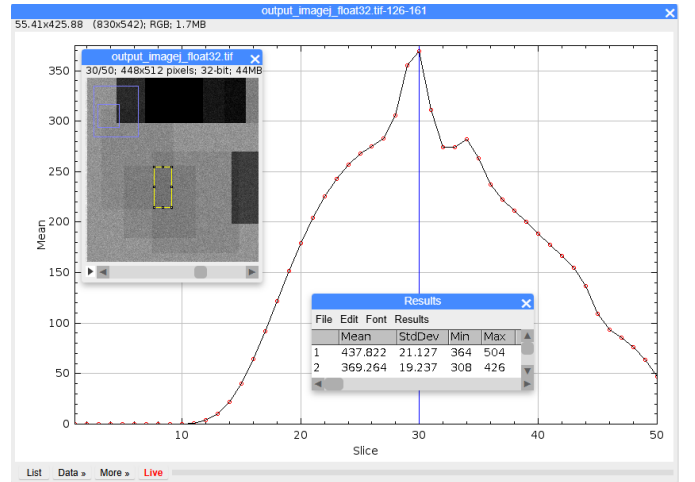
(a) Counts on a Small ROI, $SNR = 20.298$



(b) Counts on a Cobalt Blue ROI, $SNR = 19.867$



(c) Counts on a Cobalt Violet ROI, $SNR = 20.682$



(d) Counts on a ROI of three pigments superposition

Figure 10: Cobalt-based pigments: the superposition of the three samples shows a $SNR = 19.196$ but a (maximum) $CNR = 3.245$ with respect to white lead

Output image analysis: attenuation

Attenuation stacks represent the total value $-\ln(I/I_0) = \sum \mu(E)x$ basing on Lambert-Beer law, with I_0 and I being the incident and the transmitted radiation intensity, and with x and $\mu(E)$ being the total width of the sample and its linear attenuation coefficient. Both I and I_0 can be assumed to be Poisson-distributed: the following outputs display Poisson- I with details divided pixelwise by an independent Poisson- I_0 flat-field without details (zero-count pixels are floored to 1 count by default to avoid divergences).

Wood

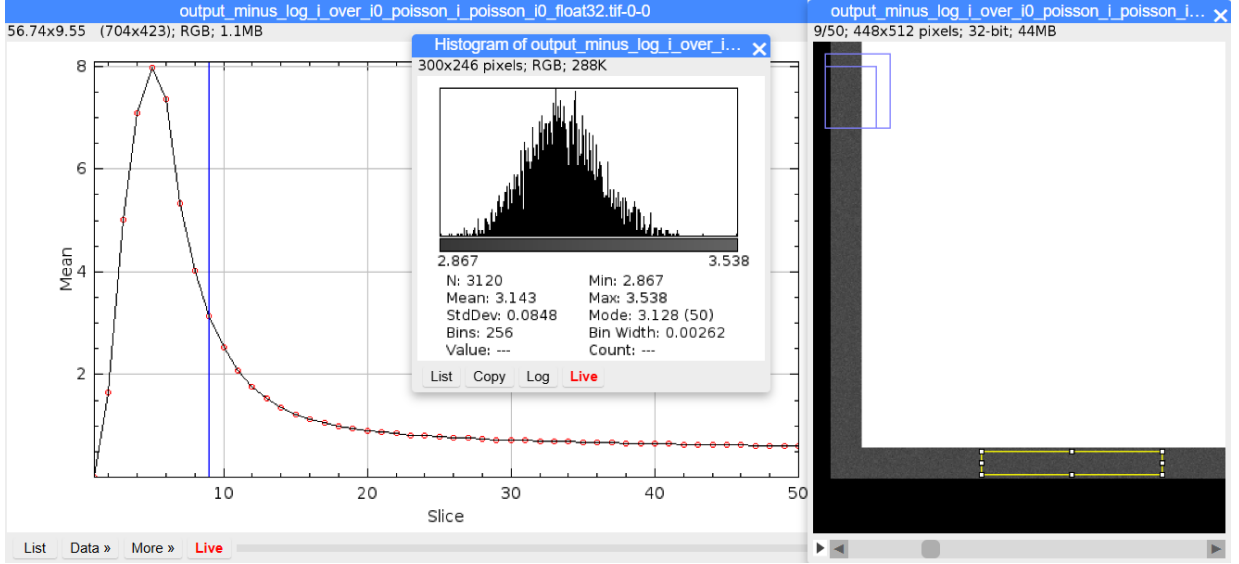


Figure 11: Total attenuation on wood: poissonian I and I_0 , $SNR = 37.064$

White lead

The lead's K-edge can be recognised: estimated $\mu(E = 89 \text{ keV}) \cdot (0.04 \text{ cm}) = 1.765 \Rightarrow \mu(E = 89 \text{ keV}) = 44.125 \text{ cm}^{-1}$, expected K-edge $\mu(88.005 \text{ keV}) = \rho_{Pb} \cdot (\mu/\rho)_{Pb} = 11.350 \cdot 7.683 \text{ cm}^{-1} = 87,202 \text{ cm}^{-1} \Rightarrow$ compatible results.

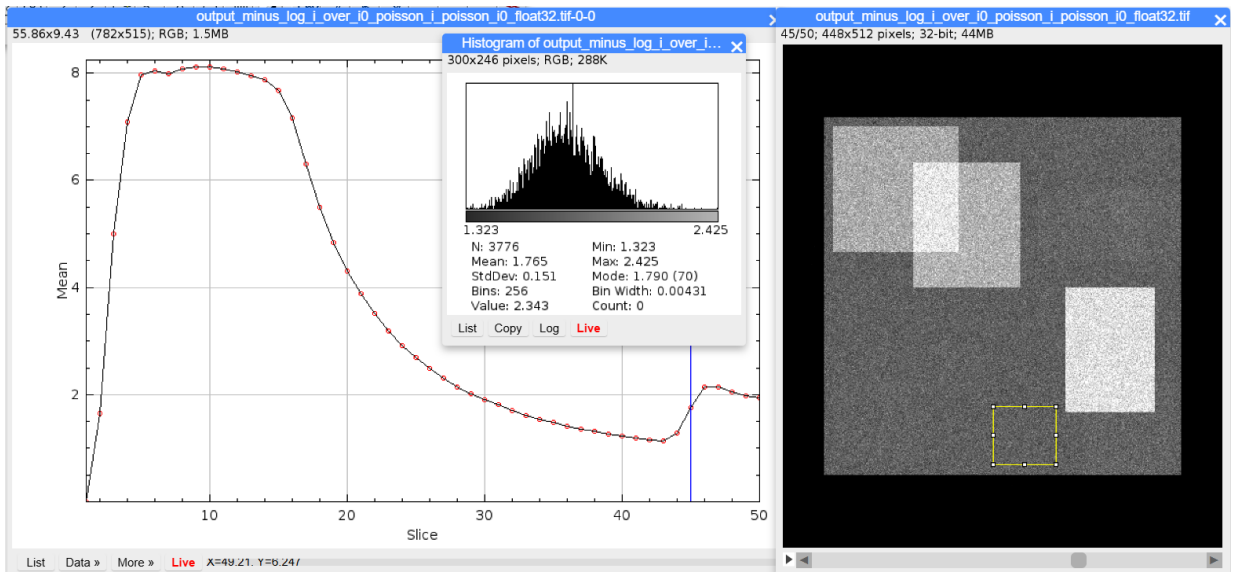


Figure 12: Total attenuation on white lead: poissonian I and I_0 , $SNR = 11.689$

Copper-based pigments

Copper-based pigments cannot be identified basing on K-edge analysis, as copper's K-edge is expected at $E = 8.979 \text{ keV}$. On the other hand, the lead's K-edge due to the presence of a lead carbonate background could still be recognised in $-\ln(I/I_0)$ spectra.

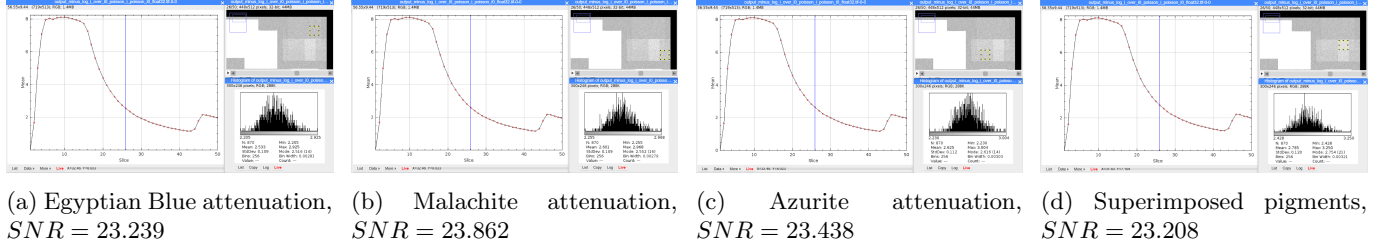
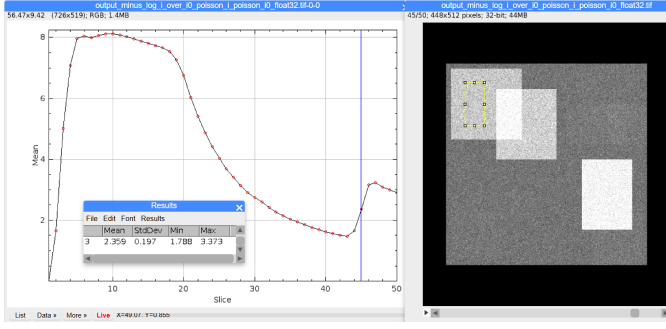


Figure 13: Copper-based pigments show similar $-\ln(I/I_0)$ spectra, strongly related to the presence of “Biacca”

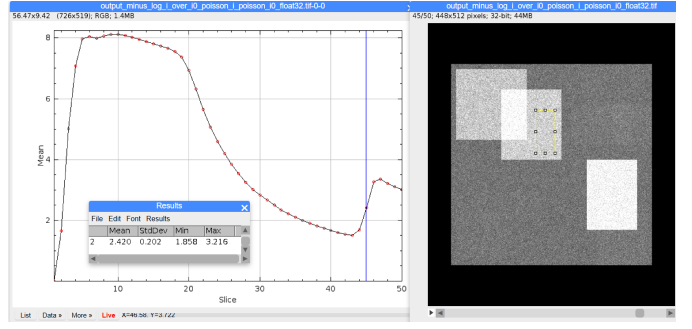
Lead-based pigments and Cinnabar

The lead's and the mercury's K-edges can be recognised and possibly distinguished (assuming instrumental resolution to be smaller than $FWHM \simeq 5 \text{ keV}$ and the white lead layer to be thin enough to avoid prevailing in the Cinnabar attenuation). Respectively:

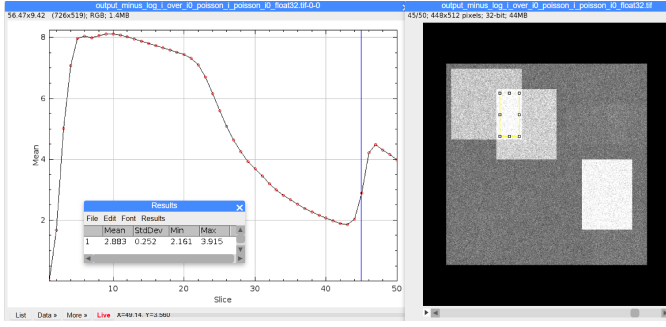
- Lead in Minium: estimated $\mu(E = 89 \text{ keV}) \cdot (0.06 \text{ cm}) = 2.359 \Rightarrow \mu(E = 89 \text{ keV}) = 39.317 \text{ cm}^{-1}$, expected K-edge $\mu(88.005 \text{ keV}) = 87,202 \text{ cm}^{-1} \Rightarrow$ compatible results
- Lead in Massicot: estimated $\mu(E = 89 \text{ keV}) \cdot (0.06 \text{ cm}) = 2.420 \Rightarrow \mu(E = 89 \text{ keV}) = 40.333 \text{ cm}^{-1}$, expected K-edge $\mu(88.005 \text{ keV}) = 87,202 \text{ cm}^{-1} \Rightarrow$ compatible results
- Red lead-based pigments superimposed: estimated $\mu(E = 89 \text{ keV}) \cdot (0.08 \text{ cm}) = 2.883 \Rightarrow \mu(E = 89 \text{ keV}) = 36.038 \text{ cm}^{-1}$, expected K-edge $\mu(88.005 \text{ keV}) = 87,202 \text{ cm}^{-1} \Rightarrow$ compatible results
- Mercury in Cinnabar: estimated $\sum \mu(E = 85 \text{ keV}) \cdot x = 1.939$, mercury's expected K-edge $\mu(83.102 \text{ keV}) = \rho_{Hg} \cdot (\mu/\rho)_{Hg} = 13.550 \cdot 8.464 \text{ cm}^{-1} = 114,687 \text{ cm}^{-1} \Rightarrow$ expected $\mu_{Hg}(E = 83.102 \text{ keV}) \cdot (0.02 \text{ cm}) + \mu_{Pb}(E = 80.000 \text{ keV}) \cdot (0.04 \text{ cm}) = 2.293 + 1.098 = 3.391 \Rightarrow$ compatible results
- Lead Yellow: estimated $\mu(E = 89 \text{ keV}) \cdot (0.03 \text{ cm}) = 1.409 \Rightarrow \mu(E = 89 \text{ keV}) = 49.967 \text{ cm}^{-1}$, expected K-edge $\mu(88.005 \text{ keV}) = 87,202 \text{ cm}^{-1} \Rightarrow$ compatible results
- Lead Yellow and Naples Yellow superimposed: estimated $\mu(E = 89 \text{ keV}) \cdot (0.05 \text{ cm}) = 1.981 \Rightarrow \mu(E = 89 \text{ keV}) = 39.620 \text{ cm}^{-1}$, expected K-edge $\mu(88.005 \text{ keV}) = 87,202 \text{ cm}^{-1} \Rightarrow$ compatible results



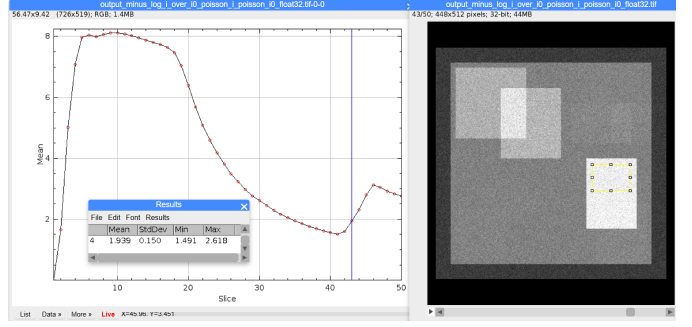
(a) Minium attenuation, $SNR = 11.974$ at $E = 89$ keV



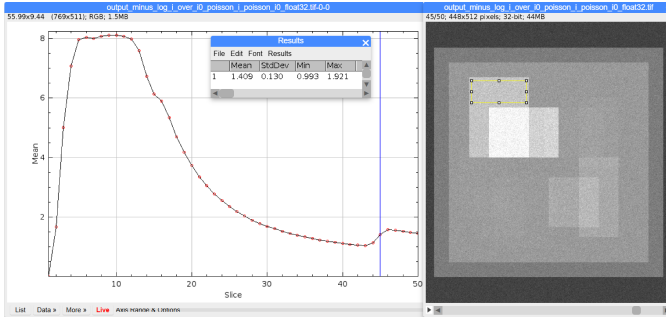
(b) Massicot attenuation, $SNR = 11.980$ at $E = 89$ keV



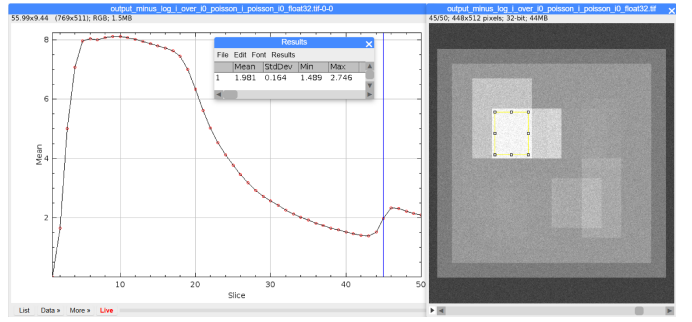
(c) Lead-based pigments superimposed, $SNR = 11.440$



(d) Cinnabar attenuation, $SNR = 12.923$ at $E = 85$ keV



(e) Lead Yellow attenuation, $SNR = 11.440$ at $E = 89$ keV; tin presence is not evident from total attenuation



(f) Lead Yellow and Naples Yellow superimposed, $SNR = 12.079$

Figure 14: Lead-based pigments and mercury-based pigment showing their K-edges

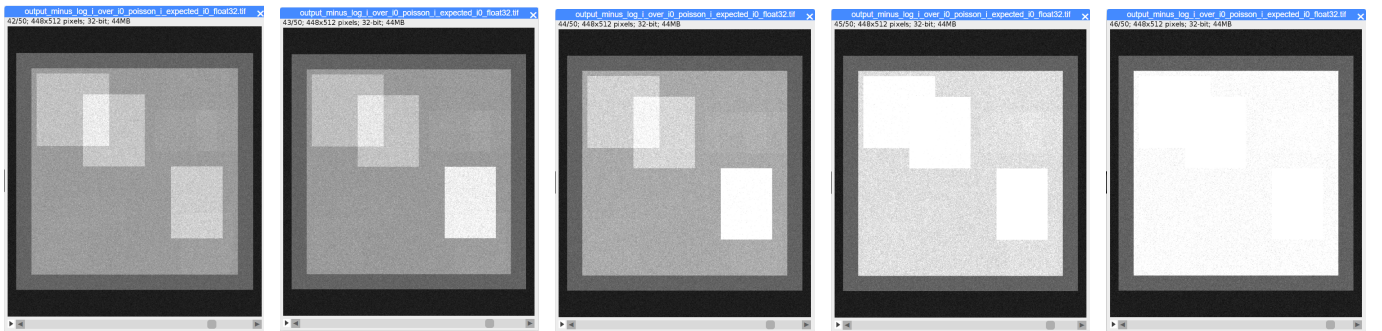
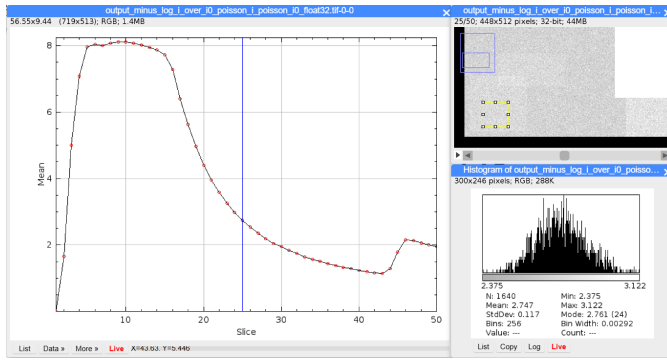


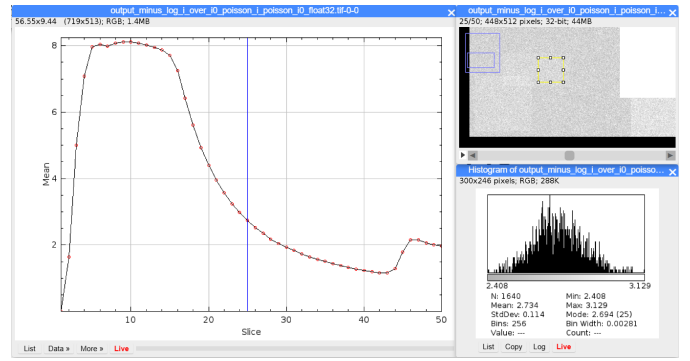
Figure 15: Total attenuation $-\ln(I/I_0)$ spectra with poissonian I and ideal I_0 , from stack 42 to stack 46 (i.e. from $E = 83$ keV to $E = 91$ keV); as expected from K-edges values of Pb and Hg, Cinnabar's attenuation peak starts at slightly lower energies than lead-based pigments' attenuation peaks

Iron-based pigments

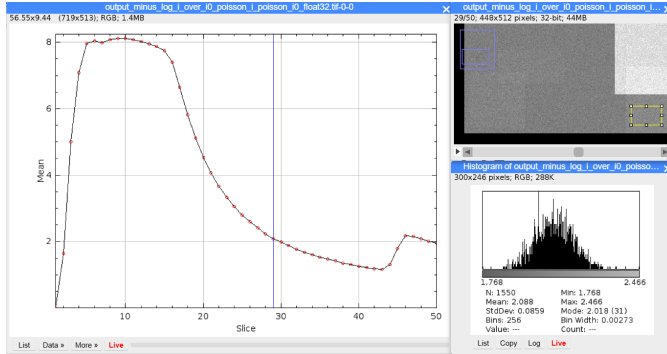
Similarly to copper-based pigments, the iron-based ones cannot be identified basing on K-edge analysis, as iron's K-edge is expected at $E = 7.112$ keV. On the other hand, the lead's K-edge due to the presence of a lead carbonate background could still be recognised in $-\ln(I/I_0)$ spectra.



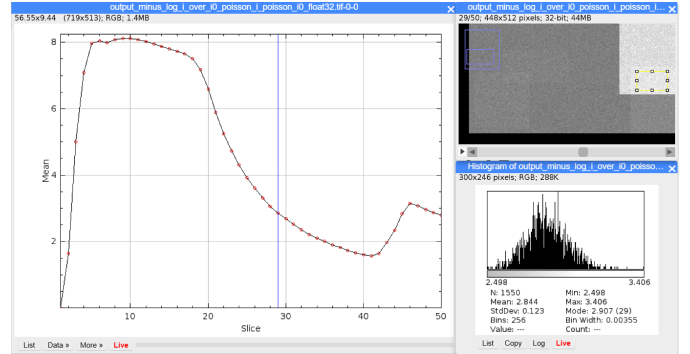
(a) Iron Oxide Yellow attenuation, $SNR = 23.479$



(b) Dark Ochre attenuation $SNR = 23.982$



(c) Hematite (Pompeii Red) attenuation, $SNR = 24.307$

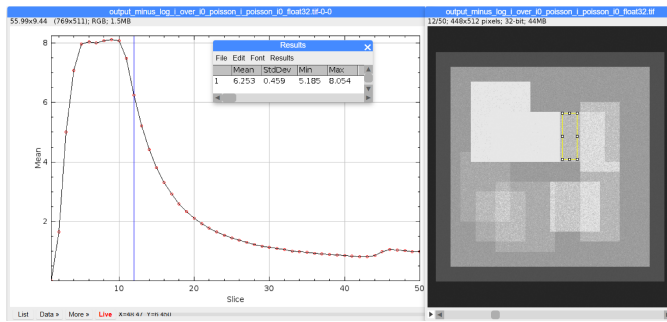


(d) Superposition of Hematite and Cinnabar, $SNR = 23.122$

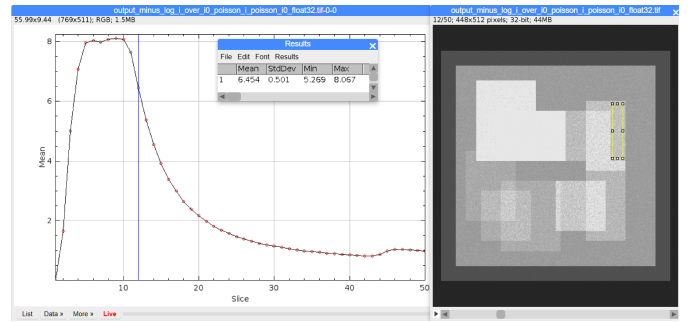
Figure 16: Iron-based pigments show similar $-\ln(I/I_0)$ spectra, strongly related to the presence of “Biacca”; as shown in figure (d), superposition with Cinnabar can also influence the attenuation peak

Arsenic-based pigments

Arsenic-based pigments cannot be identified with K-edge analysis, as arsenic’s L-edge is expected at $E = 11.867 \text{ keV}$.



(a) Orpiment attenuation, $SNR = 13.623$



(b) Realgar attenuation, $SNR = 12.882$

Figure 17: Arsenic-based pigments show similar $-\ln(I/I_0)$ spectra, strongly related to the presence of “Biacca”

Cadmium-based pigments

With an appropriate source filtration and a sufficient instrumental resolution, cadmium-based pigments might be identified with K-edge analysis, as long as the white lead layer is not particularly thick.

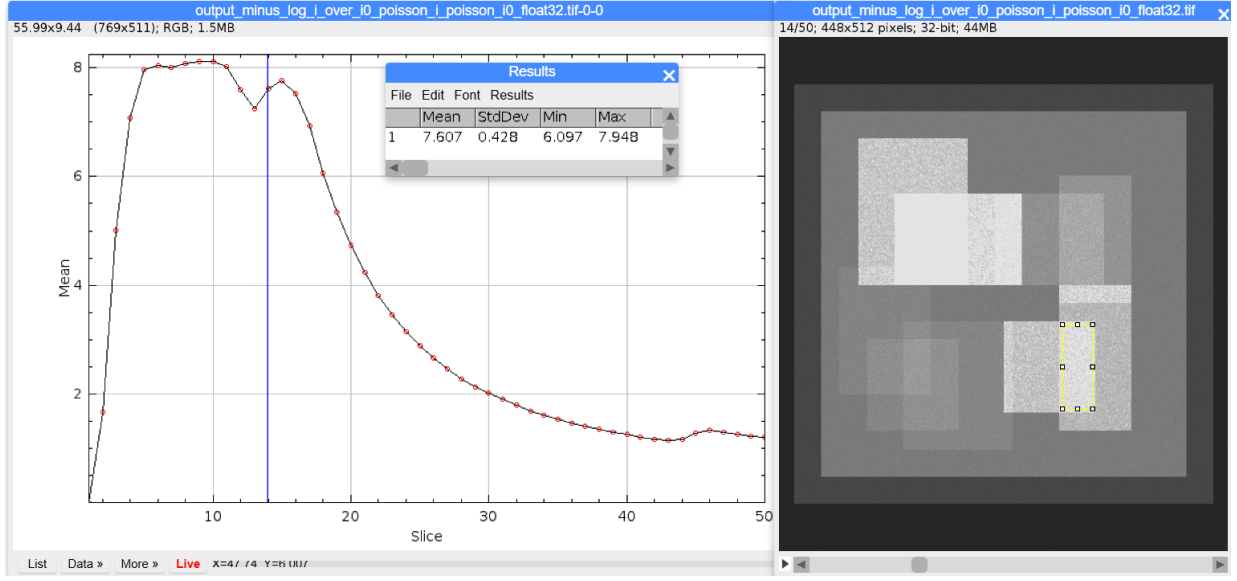


Figure 18: Total attenuation on superimposed Cadmium Yellow and Red: expected cadmium's K-edge at $E = 26.711 \text{ keV}$ can be observed ($SNR = 17.773$)

Cobalt-based pigments

As well as iron-based pigments, the cobalt-based ones cannot be identified using K-edge analysis, as cobalt's K-edge is expected at $E = 7.709 \text{ keV}$.

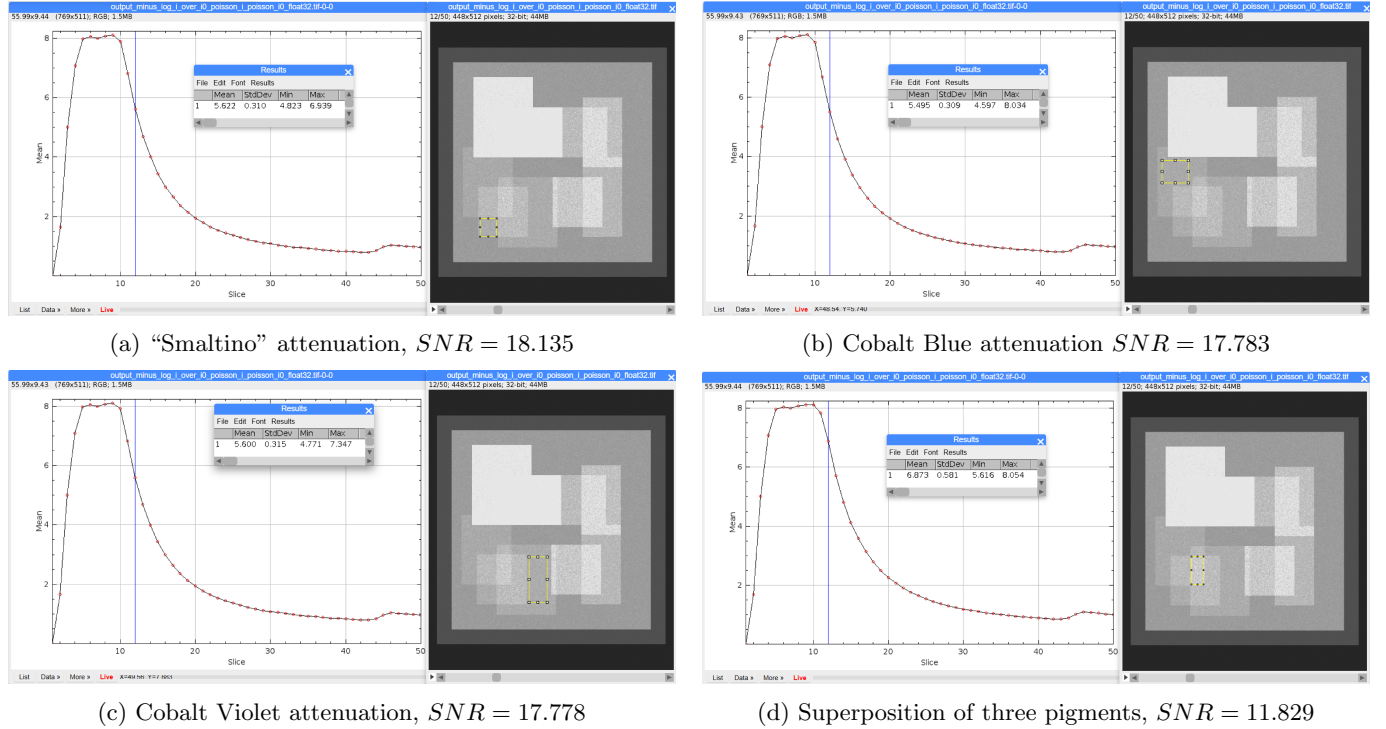


Figure 19: Total attenuation of cobalt-based pigments

K-edge subtraction analysis

Processing total attenuation output images with K-edge subtraction might enhance the probability of distinguishing lead-based pigments from mercury-based pigments. This procedure consists in duplicating the $-\ln(I/I_0)$ spectrum, deleting the first stack from a copy and the last stack from the other copy, then subtracting the so obtained images. A spectrum of the resulting image can be visualised and used to identify variations in attenuation for energy values of interest (specifically, in the range $\sim [82 \text{ keV}, 90 \text{ keV}]$).

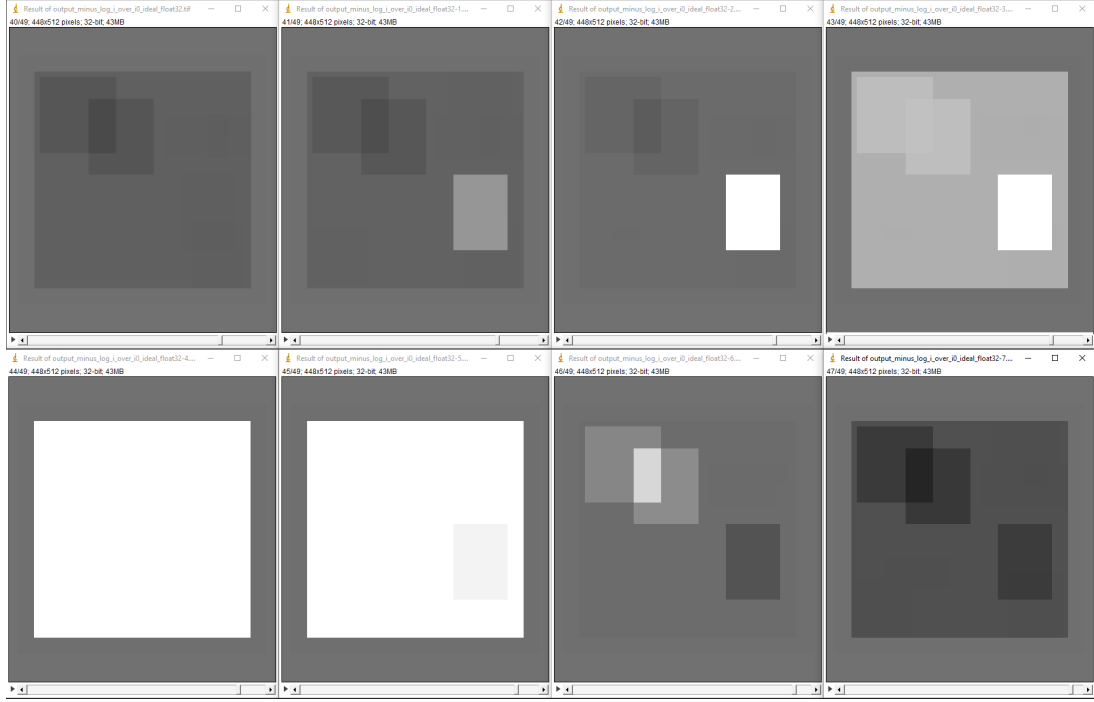
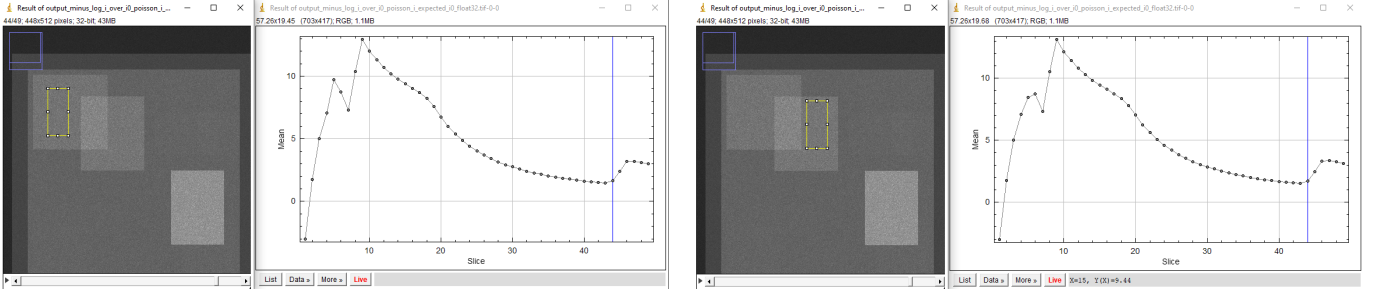
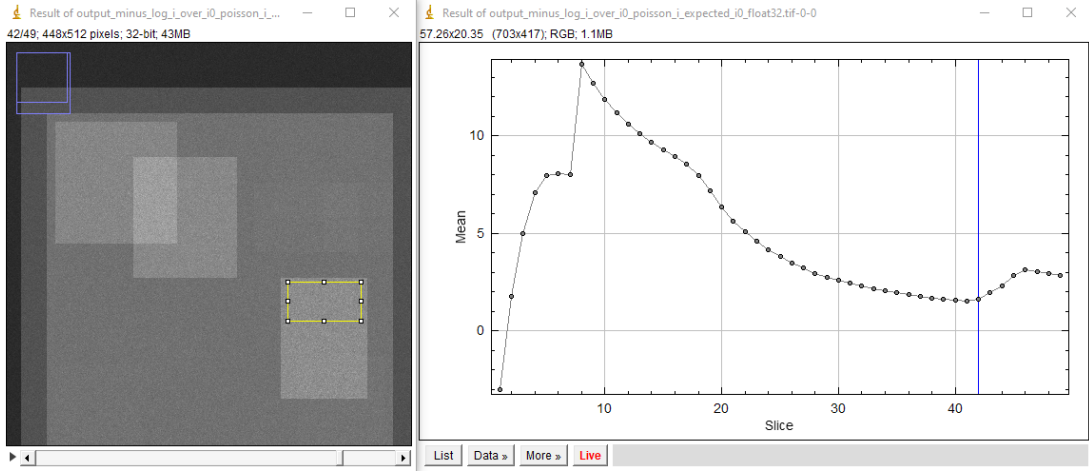


Figure 20: Stack progression from the KES image (with ideal I and I_0 in total attenuation): Minium and Massicot seem to be distinguishable from Cinnabar basing on brightness variations from stack 40 to stack 47



(a) Minium, stack 44 (i.e. difference between attenuation at $E = 89 \text{ keV}$ and at $E = 87 \text{ keV}$)

(b) Massicot, stack 44 (i.e. difference between attenuation at $E = 89 \text{ keV}$ and at $E = 87 \text{ keV}$)



(c) Cinnabar, stack 42 (i.e. difference between attenuation at $E = 85 \text{ keV}$ and at $E = 83 \text{ keV}$)

Figure 21: K-edge subtraction images

Conclusions

Results of this simulation lead to expect no evident distinction between pigments obtained from the same metal, whereas pigments belonging to the same class but based on different metals might be identifiable, depending on their composition and thickness (e.g. a lead-based red should result in a different spectrum with respect to an iron-based red or to a mercury-based red). It seems that mixtures which contain low-Z metals (such as iron or copper) cannot be identified using such an X-ray source (tungsten anode) and with such a resolution - at least, K-edges cannot be exploited as markers.

For what concerns background, wood does not seem to be absorbing a relevant quantity of incident photons, even when particularly thick, as it is a low-density organic material; lead carbonate used to cover its surface in paintings, instead, could cover some pigment signal, preventing their identification, especially when such inorganic matrix is thicker than the superimposed pigment itself.

It is crucial to take into account that this simulation overestimates experimental efficiency (as it is treated as independent of energy), materials homogeneity and possibly even experimental resolution: therefore, experimental measurements to be presented further on are expected to produce less accurate results.

Links

- [1] SpekPy documentation from Python [here](#)
- [2] Tables of linear attenuation coefficients and material densities from NIST [here](#)
- [3] Material K-edges can be found in the X-ray data booklet [here](#)